10.1. INTRODUCTION TO IMMUNOLOGY

"A healthy life is a wealthy life" and "Prevention is better than cure" are the proverbs frequently used to define the status of human health which is closely related to immunity and the immune system. Immunity is defined as the state of resistance to disease caused by specific microorganism or their toxic products. The immune system is the system of specialized cells and organs that protect an organism from diseases and infectious organisms. It is also called as the host defense system. Immunology is the study of all aspects of the immune system in an organism.

The concept of immunity can be traced back to 430BC, when Thucydides observed that individuals who had recovered from the plague would not get it a second time and could nurse other affected patients. The earliest recognised and written evidence of inducing immunity was practiced by the Chinese and Turks in the 15th Century. They either inhaled the dried crusts derived from small pox pustules or inserted them into wounds in the skin. Lady M.W. Montagu, the wife of the British ambassador to Constantinople, applied this technique on her own children and found the results to be positive. In 1798, a vaccine (from the Latin word 'vacca' , meaning 'cow') for small pox was developed by Edward Jenner. He inoculated an 8-year old boy James Phipps with material obtained from a cowpox lesion. The results were conclusive of prevention of small pox infection. Friedrich Henle was the first to discover that germs caused disease and the isolation of infectious bacteria was done by his pupil Robert Koch. Louis Pasteur developed vaccine for chicken cholera, anthrax and rabies. Modern immunology begins with the research of Metchnikoff, who discovered the phenomenon of phagocytosis in starfish and extrapolated it to macrophages in humans as cells that engulf infectious agents.



Variolation was the method used to immunize individuals against smallpox by infecting them with substance from the pustules of patients.

10.2. INFECTION

Infection is defined as the invasion and multiplication of pathogenic organisms in the host. An infection without symptoms is represented as subclinical; and with symptoms it is represented as clinically apparent. Illness caused through pathogenic organisms is termed as infectious disease or communicable disease or transmissible disease.

Epidemiology

Epidemiology is the branch of medical science that deals with the geographical distribution and timing of infectious disease occurrences. The study also includes the modes of transmission and maintenance in nature, with the goal of recognizing and controlling outbreaks.

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